The Impact of Poverty on Mortality During the First Wave of the Pandemic

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Abstract

This study aims to provide evidence that the per capita income of a specific microregion influenced mortality rates during the 2020 pandemic. We utilized national mortality data at the microregional level in Brazil, examining periods before, during, and after the peak of the initial wave of the COVID-19 pandemic. It is hypothesized that microregions situated in the lower spectrum of the per capita income distribution will exhibit higher excess mortality compared to their counterparts in the upper spectrum. This observation remains consistent even after accounting for factors such as the population aged above 70, infrastructure variables, unemployment rates, and fixed state effects.

Keywords: Income, Mortality, Inequality, COVID-19.

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1 Introduction

As of the writing of this document, November 20, 2021, the health crisis caused by the COVID-19 pandemic resulted in the death of over 600,000 Brazilians. However, little is understood about the potential heterogeneous impacts caused by differences between municipalities and microregions, particularly regarding mortality rates.

In this study, the effects of the pandemic are explored within the Brazilian context, where authorities were negligent from the onset of the crisis. This allowed the pandemic to unfold more severely in Brazil compared to other countries. No nationwide or state-wide lockdown measures were implemented, with records of only one municipality declaring a lockdown. President Jair Bolsonaro argued before the Federal Supreme Court that governors and mayors had no authority to prohibit gatherings. In São Paulo, the largest city in Latin America, mandatory mask usage was only implemented in July 2020, three months after the initial outbreak.

Prominent supply chain issues occurred. In December 2020, the city of Manaus faced a crisis due to an oxygen shortage, shortly after discovering a new, more contagious virus variant (Alpha). This variant was also linked to higher mortality rates compared to the original strain. The collapse of Manaus's health system inadvertently spread the variant to other states, necessitating the transfer of hospitalized patients who could no longer be accommodated or treated in the city.

Brazil's experience during the pandemic was a confluence of unfortunate events: every conceivable issue manifested. Authorities were slow to acknowledge the pandemic, and mayors and governors delayed containment measures. The Ministry of Health not only declined vaccine purchases for months but also promoted and funded "early treatment" drugs with no proven efficacy against the virus. The president consistently opposed any pandemic containment measures.

Given the crisis, workers without the means or infrastructure to adapt to remote work were severely impacted by the pandemic, a trait commonly associated with those having lower education levels and belonging to ethnic minorities. Evidence suggests that these groups were the first to be affected in terms of income loss and unemployment. In a country like Brazil, with a substantial informal job sector, such a shock likely pushed many into informal employment, characterized by jobs with a high risk of COVID-19 transmission.

Furthermore, as previously discussed, the lack of medical resources appears to correlate with higher mortality rates. While not as severe as Manaus, many cities across Brazil faced similar challenges. Health systems reached their limits, necessitating the expansion of bed capacities through makeshift hospitals, the transfer of patients to other states, increased demand for medical supplies, and, sadly, numerous reports of individuals dying while waiting for care.

Considering the quasi-experimental scenario arising from the exogenous shock of heterogeneous per capita income distribution during the pandemic, this study employs a difference-in-differences model. The objective is to identify the causal impact of COVID-19 on the mortality rate per 100,000 inhabitants in poorer microregions. Mortality trends over the months might have been influenced by factors not directly related to the pandemic, such as the effects of adverse seasonal diseases. To address such concerns, fixed effects for time and states will be added as controls. This will capture any invariant effects, like the influence of cultural habits inherent to the Brazilian population or potential impacts of each region's climatic characteristics.

This research combines a range of administrative data available from various government agencies to estimate the relationship between COVID-19 mortality and relative poverty at the microregional level, and to analyze potential impacts. To my knowledge, this is one of the few studies proposing to estimate the impact of regional poverty on mortality.

The structure of this paper is as follows. The first section introduces the study, explaining its significance and methodology. The second section discusses key research literature and its influence on this study. The third section details the data sources utilized, the construction of relevant variables, a descriptive analysis, and the empirical strategy used to estimate the connection between poverty and COVID-19 mortality. The fourth section presents the results of the models and a robustness analysis of these findings. Based on these results, the fifth section concludes.

2 Literature Review

The crisis spurred extensive literature on the impacts of the pandemic regarding an individual's likelihood of testing positive for COVID-19 or even undergoing testing. Using data from New York City neighborhoods, studies found that the probability of receiving a positive test result, given that one was tested, was higher in impoverished neighborhoods, areas with a higher average number of residents per household, and neighborhoods with significant populations of Black or immigrant residents. Simultaneously, impoverished neighborhoods with large immigrant populations were less likely to undergo testing (Borjas, 2020).

In the same New York City context, Almagro and Orane-Hutchinson (2020) identified racial disparities in the pandemic's impact on COVID-19 transmission, especially among Black and Latino individuals compared to their White counterparts. However, they determined that an individual's occupation had a more significant impact on the onset of COVID-19 transmission than race, especially if the job involved crowded places. Using transport sector workers as an example, a 1 percentage point increase in the number of workers in this occupation led to a 1 to 2 percentage point increase in positive COVID-19 test rates. As time progressed, this result became statistically insignificant, which the authors interpreted as evidence that the city's social distancing measures were effective in combatting virus transmission.

To understand whether minorities experienced higher mortality levels than their peers, Brandily et al. (2020) employed data from one of the most affected countries during this period to measure the pandemic's impact on excess mortality. This was constructed by averaging the previous years' data and comparing it to 2020. They employed a quasi-experimental setting to develop a "triple-differences" model that compared the evolution of excess mortality between rich and poor municipalities located in regions characterized by high or low contagion levels, between the current year and previous years. Focusing solely on municipalities within urban regions, they probed the potential role of local socio-economic determinants on the virus's infectivity. Ultimately, their aim was to estimate the causal impact of poverty on COVID-19 related mortality.

Meanwhile, other authors sought evidence on the heterogeneous impacts on the likelihood of receiving treatment, given that a person was hospitalized. Using data on Severe Acute Respiratory Syndrome (SRAG) made available by Brazil's Ministry of Health's Influenza Epidemiological Surveillance Information System (SIVEP-Gripe), Bruce et al. (2020) found that White individuals were more likely to receive intensive care compared to individuals of other races. This observation persisted even after controlling for symptoms, comorbidities, and fixed effects.

However, this disparity wasn't observed in public hospitals. The significant effect primarily stemmed from private hospitals, where only insured patients could access Intensive Care Units (ICU).

3 Data and Measurement

This study employs various datasets using microregion identifiers. In Brazil, there are 558 microregions with an average population of 373,642 inhabitants. Analysis is conducted using data released by the Brazilian Institute of Geography and Statistics (IBGE) from the 2010 Census, with 2018 population estimates. The details of other datasets are explained below.

3.1 Data Collection

To compute mortality at the microregional level, two sources are employed: DataSUS and the National Council of Health Secretaries (Conass). The data available from DataSUS pertains to deaths occurring after an individual was hospitalized in a unit of the Unified Health System. Consequently, the analysis with this data is limited to deaths recorded in public hospitals. Data from Conass originates from death records in Brazilian registries, encompassing every death that occurred on Brazilian territory and was duly registered. Conass periodically updates past data with new information on deaths, indicating a limitation in the analysis. Thus, it is vital to keep the datasets as updated as possible.

Data on COVID-19 testing is not utilized since the number of tests administered at the municipal or microregional level is not publicly available. Moreover, Brazil was notably characterized by underreporting, so test distribution is likely biased. Given the high level of COVID-19 underreporting in Brazil, the analysis employs the mortality indicator per 100,000 inhabitants - using data from Conass and DataSUS, which lack individual death causes - comparing deaths during the pandemic's peak months in 2020 against deaths in other panel months. The analysis assumes that there would be no significant changes in the number of deaths had the pandemic not occurred.

3.1.1 Pandemic

The 'pandemic' variable is a dummy representing the months with the highest number of COVID-19 cases during 2020. This period specifically covers April, May, and June, collectively known as the pandemic's first wave. (Appendix 1)

3.1.2 Mortality per 100,000 Inhabitants

The response variable is mortality per 100,000 inhabitants for each panel month. It corresponds to the number of deaths divided by the population. Therefore, the response variable is defined as:

$$Mortality_{c,m} = (Deaths_{c,m,y} / Population_c)*100$$

where $Deaths_{c,m,y}$ is the number of deaths in microregion c during month m and year y, and $Population_c$ is the number of inhabitants of microregion c.

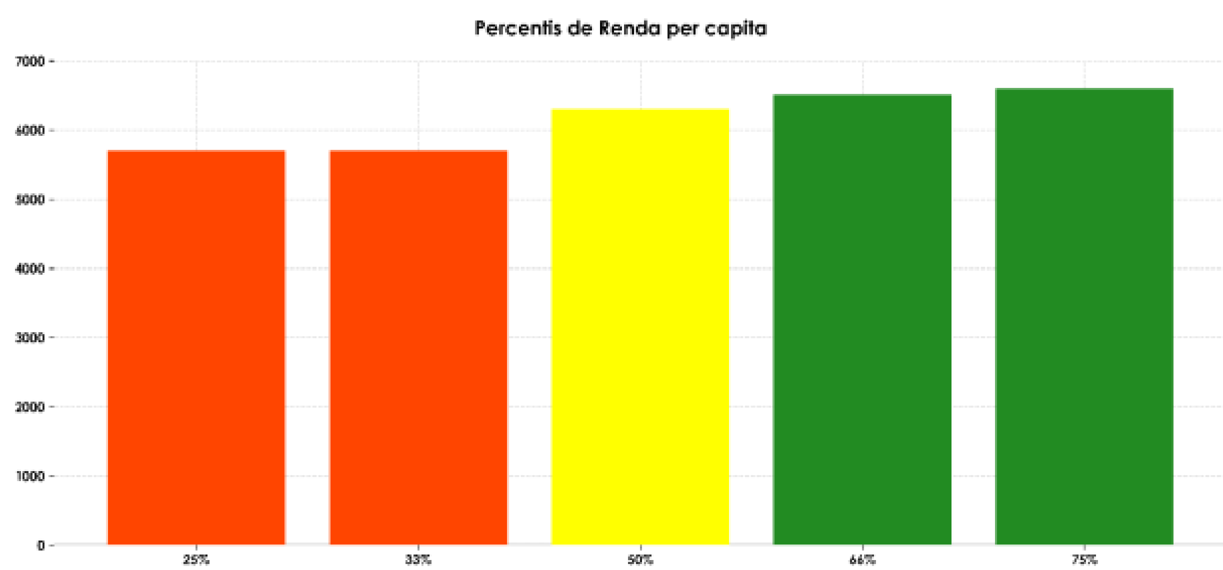
3.1.3 Poverty

The primary independent variable is a poverty measure constructed using data from the Brazilian Institute of Geography and Statistics (IBGE). Data from the 2010 Census, released every decade, is also used.

From this information, per capita income is employed to produce three poverty measures. Firstly, a dummy variable is created indicating if the microregion is below the median per capita income. Another dummy variable is constructed to denote if the microregion is in the first quartile, representing the 25% of microregions with the lowest per capita income, weighted by population. Lastly, data from the 2010 Census is used to determine the percentage of people in the microregion living with less than 50% of the minimum wage.

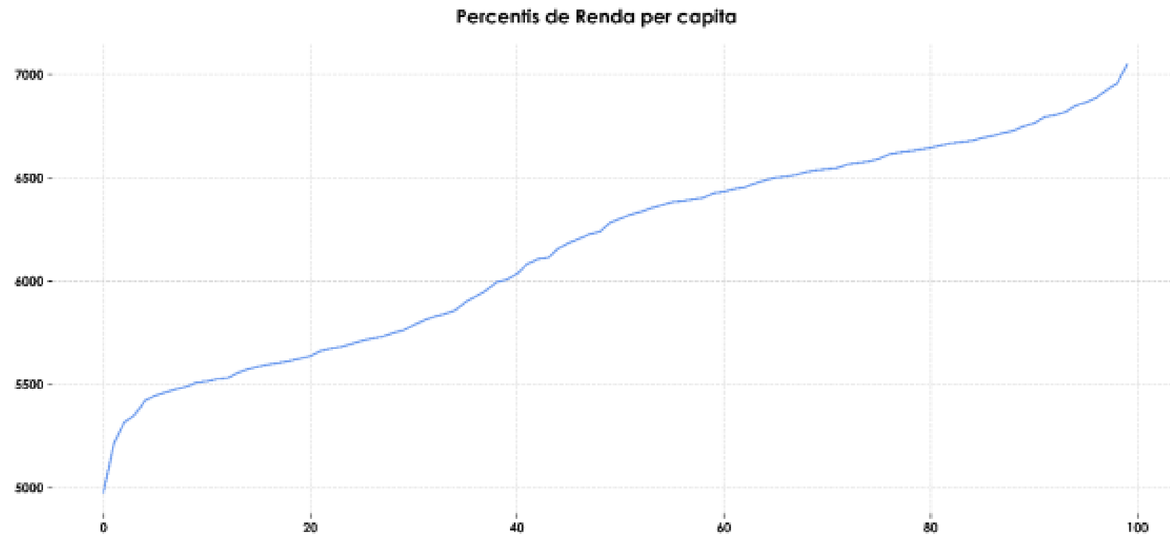
In addition to the above-mentioned variables, additional control variables are used to thoroughly isolate the potential impact of poverty.

Figure 1: Distribution of per capita income according to percentiles.



Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census, own elaboration.

Figure 2: Distribution of per capita income according to percentiles.



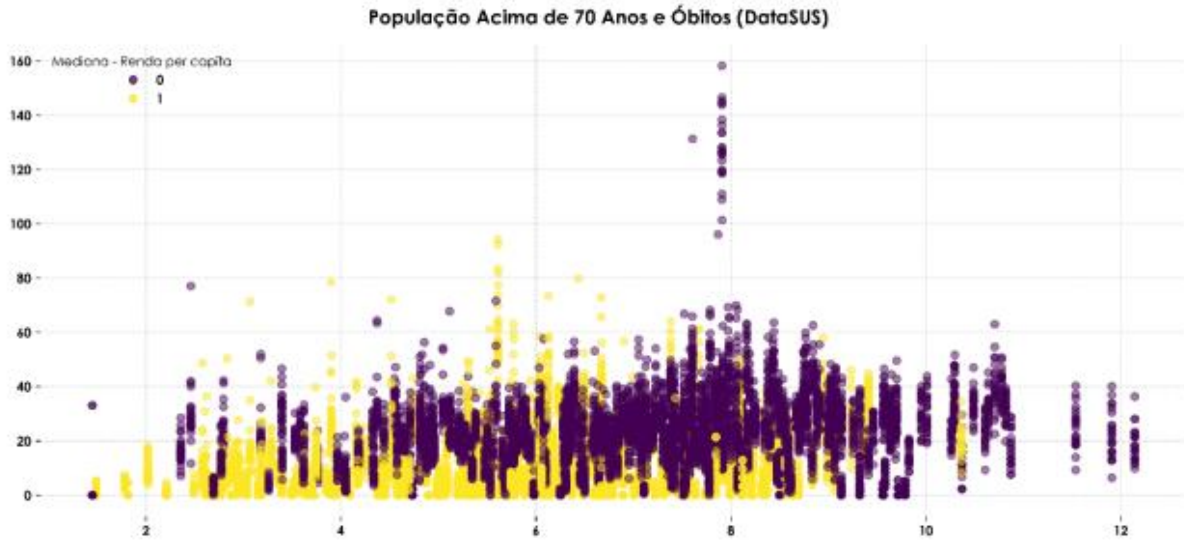
Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census, own elaboration.

3.1.4 Population and Population Density

Population: After a year of facing the pandemic, the age trend of those affected by COVID-19 is evident. Given that older individuals are more vulnerable to hospitalization, this study uses population and the population above 70 years old in each microregion as controls. Both population-related variables are expected to have positive impacts on mortality. This is because larger populations are associated with higher contagion levels, and a more significant proportion of the population above 70 years old is linked to higher death rates.

Population Density: Since COVID-19 is transmitted through airborne means, it is anticipated that microregions characterized by higher population density levels are more prone to higher contagion levels. Therefore, this variable is expected to have a positive impact on mortality per 100,000 inhabitants.

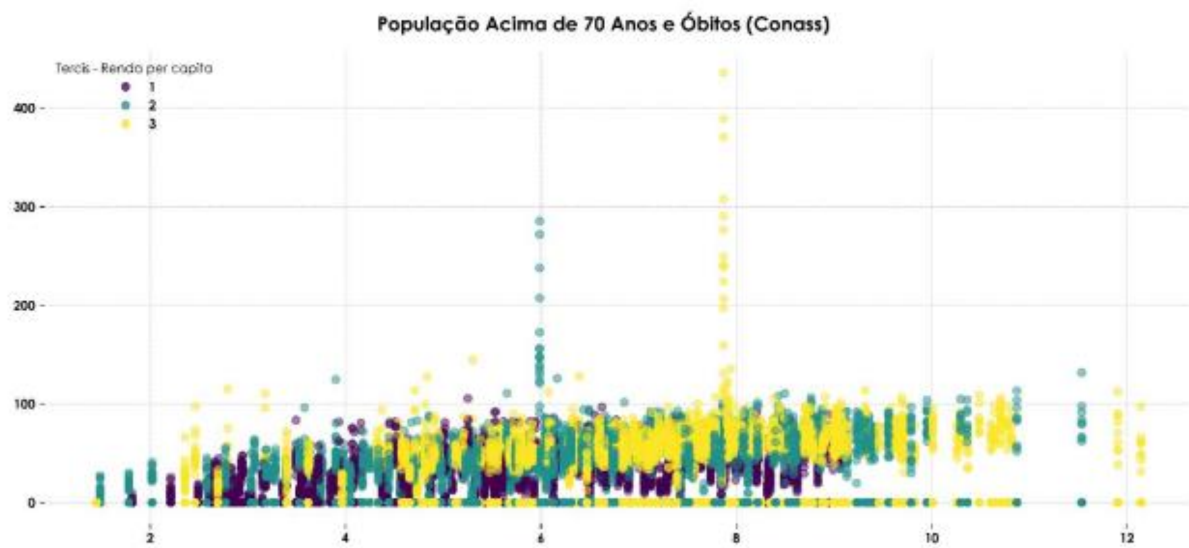
Figure 3: Distribution of hospital deaths according to the percentage of the population above 70 years old and the median per capita income.



Fonte: Instituto Brasileiro de Geografia e Estatística (IBGE) – Censo 2010 e Estimativas Populacionais de 2018,

Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census and 2018 Population Estimates, DataSUS (2019 and 2020), own elaboration.

Figure 4: Distribution of general deaths according to the percentage of the population over 70 years old and the income terciles.



Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census and 2018 Population Estimates, DataSUS (2019 and 2020), own elaboration.

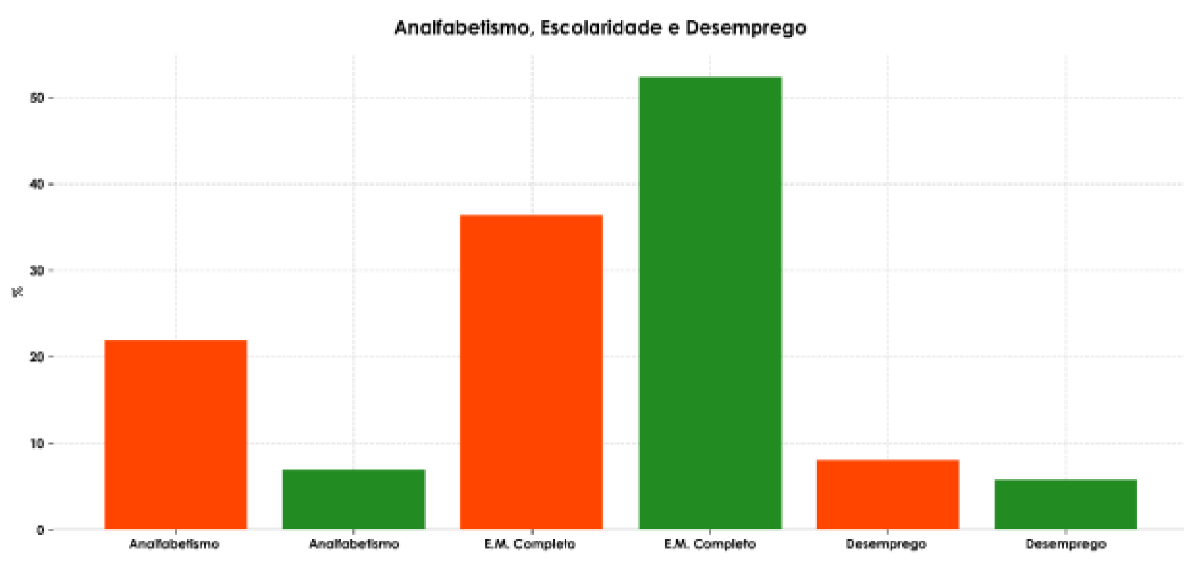
3.1.5 Illiteracy, Education, and Unemployment

Illiteracy: Given the need for containment measures and hygiene during the pandemic, people who haven't had the opportunity to become literate might face challenges in adhering to necessary guidelines. Therefore, higher illiteracy levels are expected to be positively correlated with the number of COVID-19 deaths.

Education: Reflecting the shift in the job market during the pandemic, individuals with more years of education tend to be in positions that allowed a transition from on-site to remote work, thereby reducing exposure. Thus, higher education levels should be negatively correlated with mortality during the pandemic.

Unemployment: Following the same logic as the previous variable, higher unemployment rates are expected to be negatively correlated with the number of deaths. This is based on the notion that part of the exposure to COVID-19 comes from the need to attend on-site jobs.

Figure 5: Percentage of the population in situations of illiteracy, unemployment, and completed high school education, according to the median per capita income.

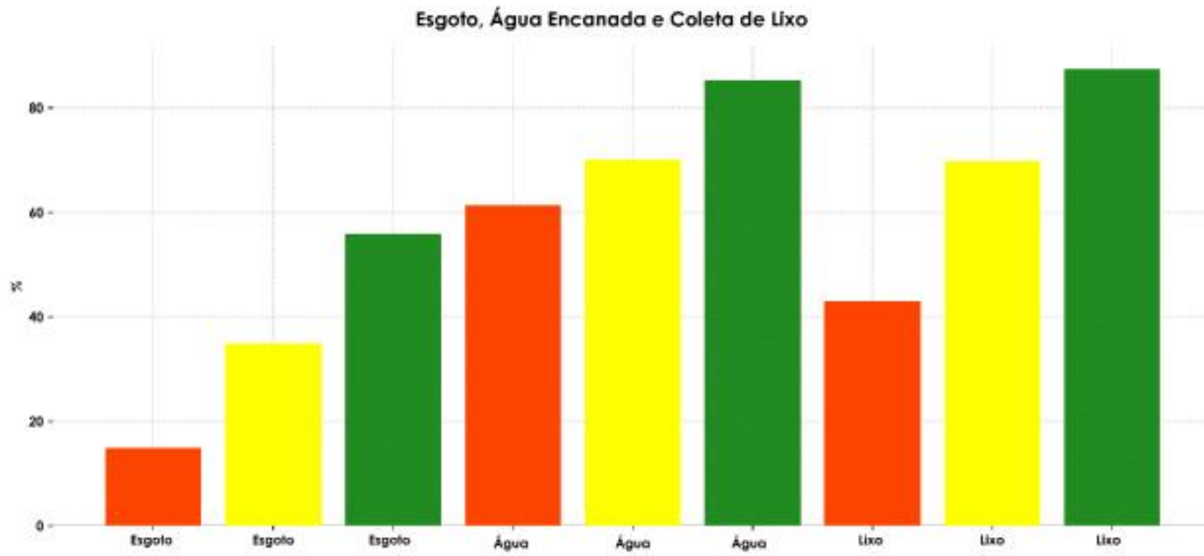


Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census, own elaboration.

3.1.6 Infrastructure: Sewage, Water, and Waste Management

The infrastructure variables describe the municipality's situation concerning basic sanitation. Since a significant part of preventing COVID-19 cases hinges on maintaining a high level of hygiene, it's anticipated that the higher the percentage of the population with access to sewage systems, piped water, and waste collection, the lower the mortality during the pandemic.

Figure 6: Percentage of the population with access to sewage systems, piped water, and waste collection, according to the per capita income terciles.



Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census, own elaboration.

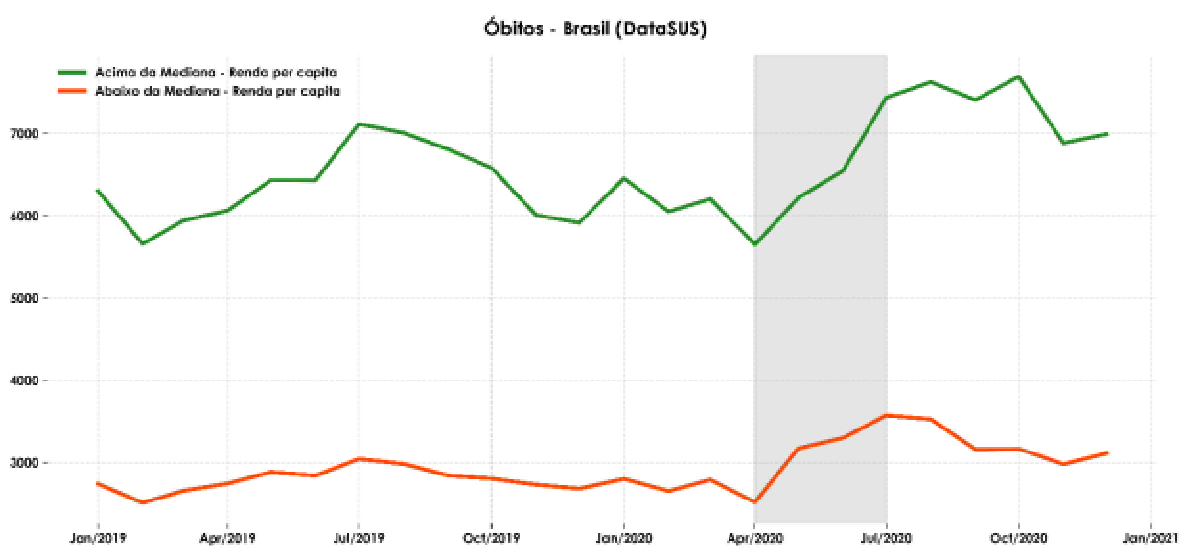
3.2 Data Processing

The data processing begins with the decision on the specification level at which the analysis is conducted. Data from Conass, IBGE, and DataSUS are at the municipal level, while data from SIVEP-Gripe are at the individual level. Given the objective of estimating the impact of poverty on mortality during the peak of the pandemic's first wave, the ideal would be to conduct the analysis at the level closest to the individual. Since most data are available only at the municipal level, this would be the most appropriate level. However, a high concentration of NaN values was observed in municipalities with low per capita income, which could bias the analysis since NaN values are disregarded during model estimation. Therefore, the analysis was conducted at the next level up from the municipal level, which is the microregional level. Additionally, due to the pandemic, the Census update was canceled. Thus, the 2010 data will be used, and for the model analysis, it will be assumed that the estimates collected have remained constant over time.

The dataset comprises a panel of data—starting from January 2019 to December 2020—where each of the 558 microregions has 24 observations, totaling 13,392 observations. For Conass data, there's a limitation as the collection is restricted to data from January to September of 2019 and 2020. The analysis using morbidity data from DataSUS consists of 3 months of treatment and 21 months of control, while the analysis with death registry data from Conass encompasses 3 months of treatment and 12 months of control.

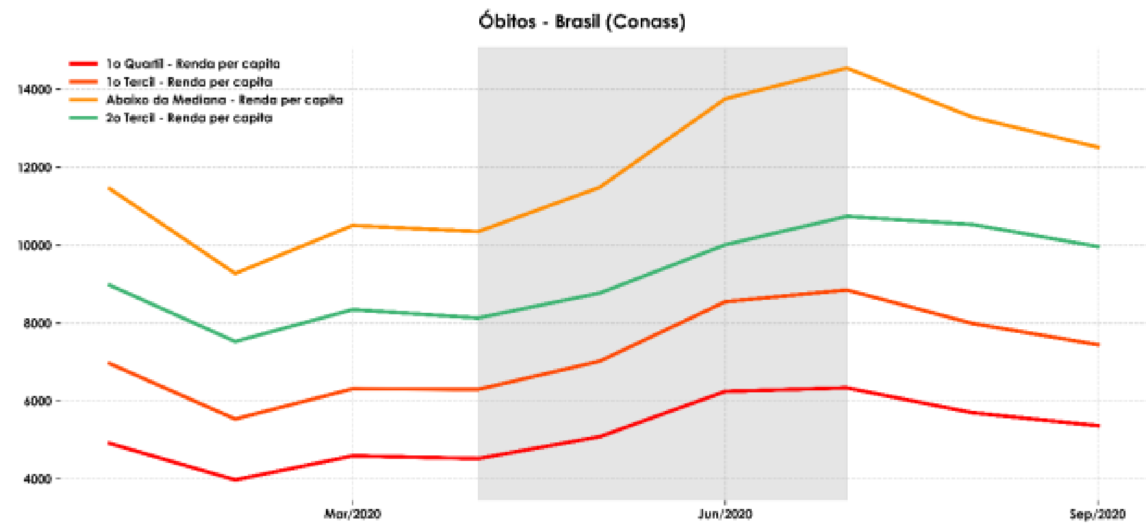
Considering the primary premise of the difference-in-differences model, it's vital to provide evidence that the control and treatment groups shared the same trend during the pre-exogenous shock period. The graph below clearly shows a similar trend between both groups of microregions. The shaded area represents the pandemic months. As the Conass dataset contains data from January to September for both 2019 and 2020, the graph has been divided into two parts, one for each year - Appendix.

Figure 7: Hospital deaths in Brazil, according to the median per capita income.



Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census, DataSUS (2019 and 2020), own elaboration.

Figure 8: General deaths in Brazil, according to the median, tercile, and quartile of per capita income.



Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census, National Council of Health Secretaries (Conass) - 2019 and 2020, own elaboration.

3.3 Empirical Strategy

The primary identification method employed is a difference-in-differences model, as represented by Equations 1 and 2. This model aims to estimate the variation in mortality across two dimensions: time and income. The first dimension involves comparing the number of deaths during the peak of the pandemic's first wave with other months, either before the pandemic's onset or after its peak. The second dimension involves categorizing the microregions based on their living standards. Two distinct methods are used for this division: those in the bottom 25% quartile of per capita income and those below the median per capita income. Intuitively, these variables aim to indicate the effect of being in a low per capita income microregion versus a high per capita income microregion. Additionally, the percentage of people living with less than 50% of the minimum wage is used to estimate the impact of an additional 1 percentage point in this measure on mortality during the pandemic.

In Equation 1, the variable $Mortality_{c,m}$ represents the mortality variables, like the ones defined in the last section.

$$Mortality_{c,m} = \beta_0 + \beta_1 * log(Income_c) + \beta_2 * Pandemic_m + \beta_3 * log(Income_c) * Pandemic_m + \varepsilon_{c,m}$$

where $Income_c$ is the income level in microregion c, $Pandemic_m$ is a dummy variable that represents the months outside of the pandemic peak with the number 0 and the months of the pandemic peak with the number 1 in microregion c and in month m, and ε is the error term in microregion c and in month m.

Another identification method employed is the difference-in-differences model with controls, represented by Equation 2.

$$Mortality_{c,m} = \beta_0 + \beta_1 * log(Income_c) + \beta_2 * Pandemic_m + \beta_3 * log(Income_c) * Pandemic_m + \beta_4 * X_{c,m} + \varepsilon_{c,m}$$

where $Income_c$ is the income level in microregion c, $Pandemic_m$ is a dummy variable that represents the months outside the pandemic peak with the number 0 and the months of the pandemic peak with the number 1 in microregion c and in month m. $X_{c,m}$ represents a control vector of microregion c and in month m, and epsilon is the error term in microregion c and in month m.

The main coefficient in both models is beta3 which represents the estimate of the difference between being in a month during the pandemic peak and being in a normal month between municipalities with low income and the others. The models in question aim to estimate the causal impact of the effect of poverty on mortality and excess mortality, considering the hypothesis that if the Covid-19 pandemic had not occurred, on average, the difference in mortality between microregions below and above the first quartile of per capita income would be constant.

4. Results

In this section, the primary findings are reported. As anticipated, the excess mortality due to the pandemic was more pronounced in low per capita income microregions when compared with medium and high per capita income microregions.

4.1 Relationship Between Poverty and Mortality

In this subsection, it's argued that the relationship between poverty and death count is causal. The difference in mortality between low-income microregions and other microregions is influenced by the per capita income pattern. The foundation of this argument rests on the premise of the difference-in-differences model, which posits that if the pandemic hadn't occurred, the mortality difference across the microregions would have remained constant. Figures 7 and 8 depict the mortality of microregions over time according to income distribution, bolstering this premise.

Estimates from the models are shown in Table 1. These estimates reveal a negative relationship between poverty and pandemic-induced mortality, with low-income microregions being most affected. The first column of Table 1 presents model estimates without control variables, using mortality per 100,000 inhabitants as calculated by Conass. During the pandemic peak, microregions below the median income had 2.51 more deaths per 100,000 inhabitants than those above the median. The second column introduces population controls, and in this specification, the impact of being below the first percentile is 2.51 deaths per 100,000 inhabitants. Finally, the fifth column reports estimates for the preferred specification, which includes all controls specified in Section 3.

In this specification, the coefficient of interest β_3 represents the excess deaths microregions experienced during the pandemic months compared to other regions, conditioned on controls, and fixed effects. Considering this model, low-income microregions experienced 2.51 additional deaths during the pandemic peak compared to the control group. This effect is statistically significant and amounts to 5.40% of the average deaths during the pandemic and 7.45 times the average deaths in the pre- and post-peak periods.

Table 2 presents estimates using excess mortality data from DataSUS. Given that DataSUS covers only public hospitals and primary care units, this analysis is limited to hospital mortality from the public health system. The first column presents model estimates using mortality without including control variables. During the pandemic, microregions below the first percentile showed a mortality rate of 2.02 per 100,000 inhabitants compared to their counterparts. The second column introduces population controls, and with this specification, the effect of being below the first percentile is an additional 2.02 deaths per 100,000 inhabitants. Finally, the fifth column presents estimates from the preferred specification, which includes all controls from Section 3.

In this model, the coefficient of interest β_3 represents the excess deaths microregions experienced during the pandemic months when compared to other regions, conditioned on controls and fixed effects. This effect is statistically significant, amounting to 12.32% of average deaths during the pandemic and 11.80 times the average deaths in the pre- and post-peak periods.

Table 1: Estimates of the difference-in-difference model for general deaths.

	Dependent Variable: Óbitos Gerais do Conass				
	Simples	Simples e Controles Populacionais	Simples e Controles de Infraestrutura	Simples e Controles A, D, E	Completo
Efeitos Fixos:	Não	Sim	Sim	Sim	Sim
Mediana	-10.277***	-4.351***	1.223**	-1.445**	0.355
	(-0.362)	(-0.523)	(-0.596)	(-0.646)	(-0.646)
Mediana*Pandemia	2.505**	2.505***	2.505***	2.505***	2.505***
	(-1.024)	(-0.879)	(-0.893)	(-0.898)	(-0.868)
Densidade		-0.002**			-0.002**
		(-0.001)			(-0.001)
População		0.00000***			0.00000***
		0			0
% Idosos		0.004			-0.261***
		(-0.084)			-0.086
% Acima de 70 Anos		3.082***			3.680***
		(-0.134)			(-0.141)
% Esgoto			0.061***		0.035***
			(-0.012)		(-0.012)
% Água			-0.030*		-0.094***
			(-0.018)		(-0.019)
% Lixo			0.143***		0.217***
			(-0.018)		(-0.02)
% Analfabetos				-0.019	-0.109*
				(-0.06)	(-0.06)
Taxa de Desemprego				-0.006	0.531***
				(-0.084)	(-0.086)
% Ensino Médio Completo				0.110***	0.014
				(-0.041)	(-0.048)
Observations	13,392	13,392	13,392	13,392	13,392
R^2	0.061	0.309	0.287	0.279	0.327
Adjusted R^2	0.06	0.306	0.284	0.276	0.324
F Statistic	436.158*** (df = 2; 13366)	186.083*** (df = 32; 13336)	173.339*** (df = 31; 13337)	166.291*** (df = 31; 13337)	170.465*** (df = 38; 13330)
Note:	*p < 0.000 ***, p < 0.001 **, p < 0.01 *, p < 0.05 '.'*				

Source: Brazilian Institute of Geography and Statistics (IBGE) - 2010 Census and Population Estimates, National Council of Health Secretaries (Conass) - 2019 and 2020, own elaboration.

Table 2: Estimates of the difference-in-difference model for hospital deaths.

	Dependent Variable: Óbitos Hospitalares do DataSUS				
	Simples	Simples e Controles Populacionais	Simples e Controles de Infraestrutura	Simples e Controles A, D, E	Completo
Efeitos Fixos:	Não	Sim	Sim	Sim	Sim
Mediana	-13.282***	-7.947***	-1.651***	-1.651***	2.061***
	(-0.225)	(-0.347)	(-0.373)	(-0.373)	(-0.4)
Mediana*Pandemia	2.018***	2.018***	2.018***	2.018***	2.018***
	(-0.635)	(-0.585)	(-0.559)	(-0.559)	(-0.538)
Densidade		-0.002***			-0.001*
		(-0.0005)			(-0.0004)
População		0.00000***			-0.00000***
		0			0
% Idosos		1.066***			0.436***
		(-0.056)			(-0.054)
% Acima de 70 Anos		1.095***			1.666***
		(-0.089)			(-0.088)
% Esgoto					0.218***
					(-0.037)
% Água					-0.068
					(-0.053)
% Lixo					0.720***
					(-0.03)
% Analfabetos			0.123***	0.123***	0.092***
			(-0.007)	(-0.007)	(-0.007)
Taxa de Desemprego			0.095***	0.095***	-0.009
			(-0.011)	(-0.011)	(-0.012)
% Ensino Médio Completo			0.175***	0.175***	0.079***
			(-0.011)	(-0.011)	(-0.012)
Observations	13,392	13,392	13,392	13,392	13,392
R^2	0.224	0.344	0.401	0.401	0.445
Adjusted R^2	0.223	0.341	0.398	0.398	0.442
F Statistic	1,928.605*** (df = 2; 13366)	218.558*** (df = 32; 13336)	287.770*** (df = 31; 13337)	287.770*** (df = 31; 13337)	280.757*** (df = 38; 13330)
Note:	*p < 0.000 ***, p < 0.001 **, p < 0.01 *, p < 0.05 '.'*				

Source: Brazilian Institute of Geography and Statistics (IBGE) - 2010 Census and Population Estimates, DataSUS - 2019 and 2020, own elaboration.

4.2 Robustness

4.2.1 Marginal Variations

To examine how the impact of poverty shifts with marginal variations, different income levels were employed to estimate the impact on mortality per 100,000 inhabitants. The main model utilized for this analysis is represented by Equation 3:

$$Mortality_{c,m} = \beta_0 + \beta_1 * log(Median_c) + \beta_2 * Pandemic_m + \beta_3 * log(Median_c) * Pandemic_m$$
$$+ \beta_4 * log(FirstQ_c) + \beta_5 * log(FirstQ_c) * Pandemic_m$$
$$+ \beta_6 * log(SecondT_c) + \beta_7 * log(SecondT_c) * Pandemic_m + \beta_8 * X_{c,m} + \varepsilon_{c,m}$$

where $Median_c$ is a dummy variable representing whether microregion c is below the median, $FirstQ_c$ is a dummy variable indicating if microregion c is within the first quartile of per capita income, $SecondT_c$ is a dummy variable indicating if microregion c is within the second tercile of per capita income, $Pandemic_m$ is a dummy variable representing the months outside the peak of the pandemic with the number 0 and the months of the pandemic peak with the number 1 in microregion c, and in month m, $X_{c,m}$ represents a control vector of microregion c, and in month m, and ε is the error term in microregion c, and in month m.

The findings are presented in Table 3, suggesting that marginal variations in per capita income increase the number of deaths per 100,000 inhabitants up to a certain threshold. Beyond this income level, the impact reverses, indicating that high-income microregions maintain fewer deaths per 100,000 inhabitants compared to their counterparts.

Contrary to expectations, for Conass data, when other income measures were added, microregions below the first income quartile showed a statistically insignificant negative impact. Meanwhile, the impact of being below the median income increased and remained significant. One potential explanation might be the heterogeneous distribution of notary offices across microregions. Low-income microregions may have fewer notary offices, leading individuals to register deaths in the nearest office, often found in medium income microregions.

For SUS hospital death data, the impact appeared as expected. Upon adding additional income variables, the impacts of being below the median or the second tercile appeared positive but statistically insignificant. In contrast, being below the first quartile showed a positive and statistically significant impact. A possible reason might be the extensive reach of the Unified Health System (SUS), ensuring data collection is as uniform as possible across microregions.

Table 3: Difference-in-difference model estimates for hospital deaths and general deaths.

	Dependent Variables:			
	Óbitos - Conass		**Óbitos - DataSUS**	
	Completo	Completo	Completo	Completo
Efeitos Fixos:	Sim	Sim	Sim	Sim
Mediana*Pandemia	2.505*** (-0.868)	3.013*** (-1.127)	2.018*** (-0.538)	1.122 (-0.695)
PrimeiroQ*Pandemia		-1.006 (-1.422)		1.782** (-0.877)
SegundoT*Pandemia		-0.27 (-1.069)		0.165 -0.659
Mediana	0.355 (-0.646)	-1.327* (-0.677)	2.061*** (-0.4)	0.595 (-0.418)
Primeiro Quartil		-4.038*** (-0.657)		-5.397*** (-0.405)
Segundo Tercil		1.133*** (-0.428)		0.202 (-0.264)
Observations	13,392	13,392	13,392	13,392
R^2	0.327	0.331	0.445	0.454
Adjusted R^2	0.324	0.328	0.442	0.451
F Statistic	170.465*** (df = 38; 13330)	160.711*** (df = 41; 13327)	280.757*** (df = 38; 13330)	263.510*** (df = 42; 13326)
Note:	*$p < 0$ ***, $p < 0.001$ **, $p < 0.01$ *, $p < 0.05$ '.'*			

Source: Brazilian Institute of Geography and Statistics (IBGE) - 2010 Census and Population Estimates, DataSUS - 2019 and 2020, own elaboration.

In addition, Table 4 examined the impacts of a microregion being in the last quartile of income - within the top 25% in terms of per capita income, considering both the absolute and marginal impact. For both general death data and hospital death data, the impact of a microregion being above the last quartile of income is negative and significant. Considering the issue of variable marginality, when adding other income variations, for the Conass data, the effect increased and remained significant, while for the DataSUS data the effect decreased but remained significant, and the effect of being within the second tercile of per capita income also proved negative and significant. This indicates that there isn't just an increase in mortality from low-income microregions; there's also a decrease in mortality from being in extremely high-income microregions.

Table 4: Estimates of the difference-in-difference model for hospital deaths and general deaths.

	Dependent Variables:			
	Óbitos - Conass		**Óbitos - DataSUS**	
	Completo	Completo	Completo	Completo
Efeitos Fixos:	Sim	Sim	Sim	Sim
ÚltimoQ*Pandemia	-3.602***	-3.708***	-2.099***	-1.921**
	(-1.001)	(-1.402)	(-0.621)	(-0.869)
Mediana*Pandemia		0.662		1.072
		(-1.111)		(-0.689)
SegundoT*Pandemia		-1.243		-1.220*
		(-1.055)		(-0.654)
Mediana		-0.645		1.342***
		(-0.672)		(-0.416)
Segundo Tercil		3.322***		2.312***
		(-0.453)		-0.281
Último Quartil	1.173**	3.470***	0.622*	2.143***
	(-0.559)	-0.651	-0.347	-0.404
Observations	13,392	13,392	13,392	13,392
R^2	0.327	0.33	0.443	0.448
Adjusted R^2	0.324	0.327	0.441	0.445
F Statistic	170.672*** (df = 38; 13330)	156.381*** (df = 42; 13326)	279.065*** (df = 38; 13330)	256.998*** (df = 42; 13326)
Note:	*p < 0 ***, p < 0.001 **, p < 0.01 *, p < 0.05 '.'*			

Source: Brazilian Institute of Geography and Statistics (IBGE) - 2010 Census and Population Estimates, DataSUS - 2019 and 2020, own elaboration.

4.2.2 Standard Error

When analyzing the standard errors of the estimated models, evidence was found suggesting the need for clustering to maintain the robustness of the standard errors. A White-style covariance matrix was used, and clustering was performed at the microregional level to ensure the robustness of the residuals.

The results are presented in Table 5, where the outcomes maintain their significance. This indicates that the estimates are robust.

Table 5: Difference-in-difference model estimates for hospital deaths and general deaths with clustered standard errors.

	Dependent Variables:			
	Óbitos - Conass		**Óbitos - DataSUS**	
	Completo	Completo	Completo	Completo
Efeitos Fixos:	Sim	Sim	Sim	Sim
Mediana*Pandemia	2.505*	3.013*	2.018***	1.122
	(-0.868)	(-1.127)	(-0.538)	(-0.695)
PrimeiroQ*Pandemia		-1.006		1.782*
		(-1.422)		(-0.877)
SegundoT*Pandemia		-0.27		0.165
		(-1.069)		(-0.659)
Mediana	0.355	-1.327*	2.061***	0.595
	(-0.646)	(-0.677)	(-0.4)	(-0.418)
Primeiro Quartil		-4.038***		-5.397***
		(-0.657)		(-0.405)
Segundo Tercil		1.133**		0.202
		(-0.428)		(-0.264)
Observations	13,392	13,392	13,392	13,392
R^2	0.327	0.331	0.445	0.454
Adjusted R^2	0.324	0.328	0.442	0.451
F Statistic	170.465*** (df = 38; 13330)	160.711*** (df = 41; 13327)	280.757*** (df = 38; 13330)	263.510*** (df = 42; 13326)
Note:	*$p < 0$ ***, $p < 0.001$ **, $p < 0.01$ *, $p < 0.05$ '.'*			

Source: Brazilian Institute of Geography and Statistics (IBGE) - 2010 Census and Population Estimates, DataSUS - 2019 and 2020, own elaboration.

5 Conclusion

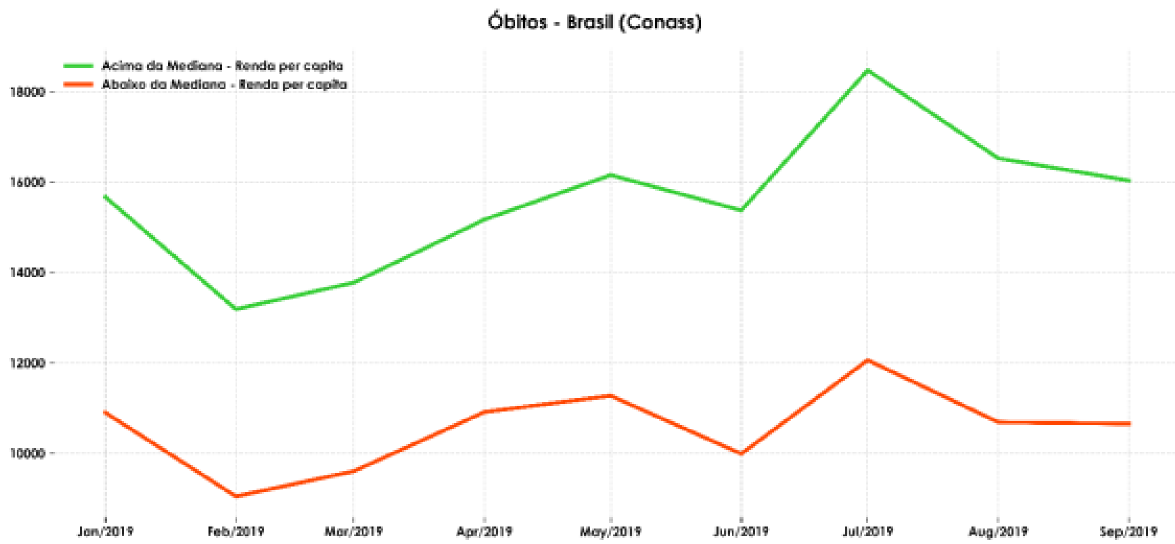
In this study, evidence was presented suggesting that in Brazil, microregions with lower per capita income experienced higher levels of excess mortality during the pandemic when compared to their counterparts with higher income levels. Leveraging the quasi-experimental nature of the pandemic, a differences-in-differences model was employed to estimate the impact of poverty on mortality during the peak months of the pandemic. The findings indicate that microregions characterized by lower per capita income experienced an increase of approximately 2 deaths per 100,000 inhabitants, more than other microregions.

From these initial results, it can be inferred that high-income microregions suffered less during the pandemic. Consequently, the pandemic's impact resulted in a decrease in mortality for high-income microregions and an increase in mortality for low-income microregions when the marginality of per capita income is considered. It's essential to highlight the need for caution when interpreting these results as causal due to the study's limitations.

It is hoped that the findings of this research can serve as a foundation or inspiration for public policies aimed at addressing the challenges created and exacerbated by the pandemic. Moreover, this work may motivate further analyses of the adverse impacts of the pandemic in the future.

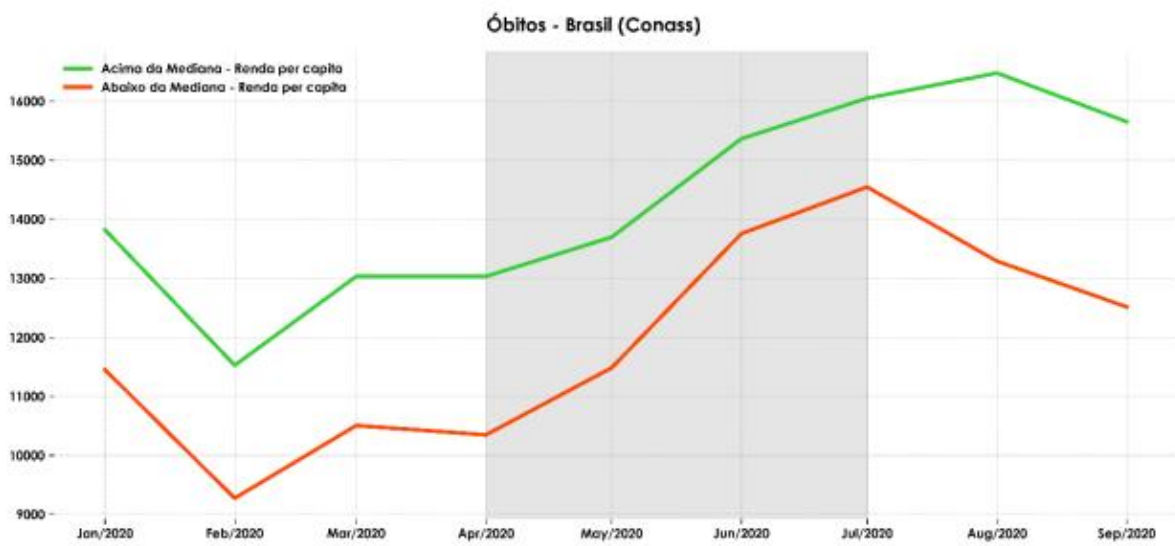
Appendix

Figure 9: Overall deaths in Brazil, according to the median per capita income.



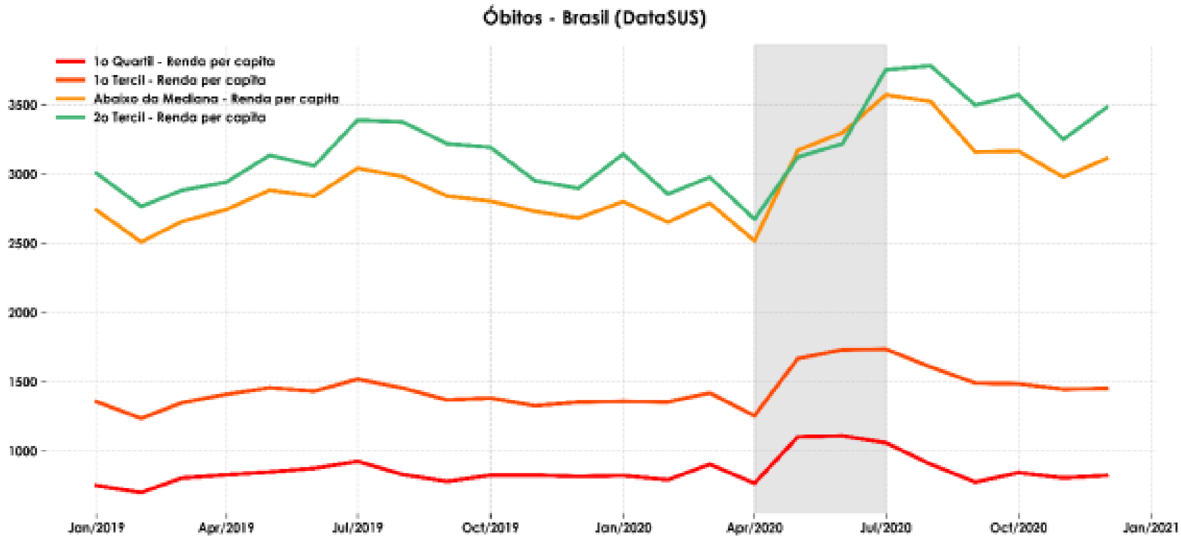
Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census, National Council of Health Secretaries (Conass) - 2019, own elaboration.

Figure 10: Overall deaths in Brazil, according to the median per capita income.



Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census, National Council of Health Secretaries (Conass) - 2019, own elaboration.

Figure 11: Hospital deaths in Brazil, according to the median, quartiles, and terciles of per capita income.



Source: Brazilian Institute of Geography and Statistics (IBGE) – 2010 Census, National Council of Health Secretaries (Conass) - 2019, own elaboration.

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Data Sources

General Morbidity Conass: Transparency Portal - Civil Registry

SUS Hospital Morbidity: SUS Hospital Morbidity (SIH/SUS) – DATASUS (saude.gov.br)

Sanitation: Sanitation – Censuses 1991, 2000, and 2010 – DATASUS (saude.gov.br)

Education: Education – Censuses 1991, 2000, and 2010 – DATASUS (saude.gov.br)

Work and Income: Work and income – Censuses 1991, 2000, and 2010 – DATASUS (saude.gov.br)

Gross Domestic Product: Gross Domestic Product – DATASUS (saude.gov.br)

GitHub

The code is available at: arthuralberti/Covid-19-Analysis (github.com)